

Future Design Direction for SRAM Data Array: Hierarchical Subarray with Active Interconnect

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Abstract—In sub 10 nm nodes, the growing dominance of interconnects in chips poses challenges in designing large-size static random-access memory (SRAM) subarrays. The main issue is the write failure problem arising from the increased resistance and capacitance for bitline (BL) and wordline (WL). To tackle this issue, the SRAM subarray design incorporates conventional (Conv.) divided WL and divided BL techniques based on 14-A-compatible (A14) nanosheet (NS) technology. This approach allows for various subarray sizes with successful write operations, resulting in improved subarray-level performance and power (PP). However, the additional logic gates come with an area penalty that may degrade the overall performance, power, and area (PPA) at the macro level due to increased inter-subarray interconnect overhead. To overcome this limitation, the active interconnect (AIC) design is proposed with the features of fabricating another or multiple active regions at the back-end of line (BEOL) layers. By moving these extra logic gates from front-end of line to BEOL in the AIC divided subarray design, the area penalty is significantly mitigated without compromising PP compared to the standard (Std.) and Conv. divided counterparts. To achieve this concept, carbon nanotube gate-all-around transistor is explored as potential BEOL-compatible device. In this research, a comprehensive design-technology co-optimization analysis is conducted to verify the value and potential benefits of up to 65% macro-level **energy-delay-area product** improvement by AIC divided subarray design compared to the Std. subarray design.

Index Terms—SRAM, Nanosheet, DTCO, active interconnect (AIC), Carbon Nanotube FET (CNFET).

I. INTRODUCTION

In interconnect-dominant nodes, the static random-access memory (SRAM) subarray area becomes a critical concern due to its impact on inter-subarray interconnect overhead. This inter-subarray interconnect, in turn, exerts significant influence over the overall macro-level performance, power, and area (PPA) [1, 2]. Fig. 1 presents an insightful graph of the energy-delay-area product (EDAP) for a 16 MB data array consisting of subarrays with varying number of row (-R) and column (-C)

based on high-density (HD) 6-transistors (6T) SRAM bitcell. As depicted in Fig. 1, a significant 56% reduction in macro-level EDAP is achieved when transitioning from a small- to a large-size standard (Std.) subarray design. Notably, these macro-level results have not yet considered the potential write failure risk (WFR) [3] within large Std. subarray. This EDAP reduction is primarily attributed to the decreased inter-subarray interconnect congestion [1]. Remarkably, given the dominance of inter-subarray interconnects, the overall EDAP continues to decrease even as subarray-level power and performance (PP) increases with bigger subarray size [1]. This observation propels cache design toward larger SRAM subarrays to attain further macro-level PPA improvements.

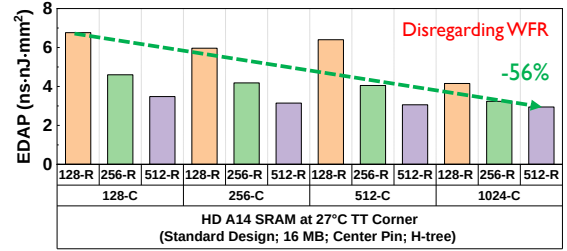


Fig. 1. Macro-level EDAP benchmark for different subarray sizes of A14 NSFET SRAM with Std. subarray design (disregarding WFR) (Fig. 3(a)), data capacity of 16 MB, **center pin access**, and H-tree technology based on typical-typical (TT) corner and nominal temperature (27°C). C: column; R: row.

A. Problem Statement

However, enabling a large-size SRAM subarray poses challenges due to the significant parasitic resistance and capacitance (RC) overhead on both bitline (BL) and wordline (WL) [1-3]. This increased RC may cause write failures (Fig. 2) in the Std. subarray design (Fig. 3(a) and Table I) [3]. Note that the write operation is considered to be at risk of failure when the write margin (WM) falls below threshold voltage (V_T), which is 300 mV in this work [3]. The scope of system-level optimization becomes limited to accommodate a wide variety of subarray sizes for diverse applications and cache configurations. Consequently, the design technology co-optimization (DTCO) lacks a complete overview of subarrays' PPA in advanced technology nodes.

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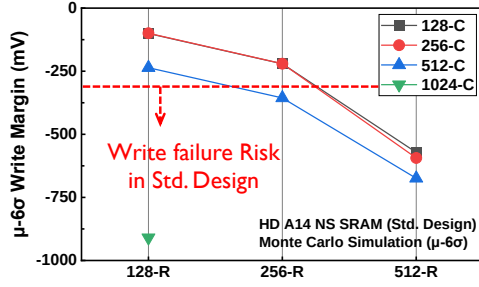


Fig. 2. Worst-case WM of HD A14 NSFET SRAM subarray with Std. design based on 6-sigma (6σ) Monte Carlo simulation at the worst-case temperature (80°C).

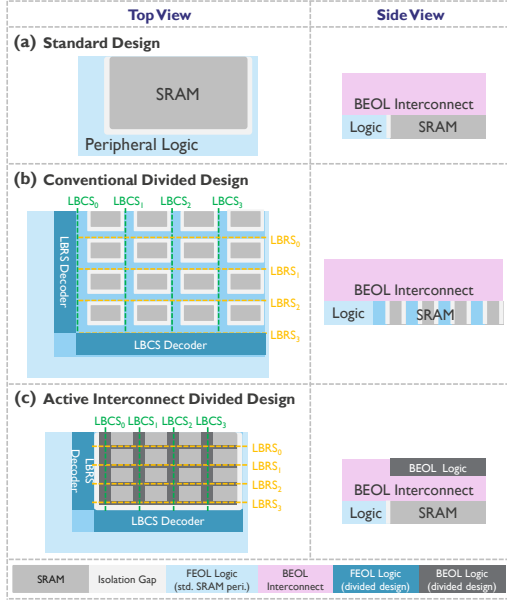


Fig. 3. Schematics of top and side views of SRAM subarray designs. (a) Standard (Std.) design: undivided SRAM bitcell array. (b) Conventional (Conv.) divided design: SRAM bitcell array is divided into several local blocks by extra FEOL logic gates. (c) Active interconnect (AIC) divided design: SRAM bitcell array is divided into several local blocks by additional BEOL logic gates.

Table I. Design types and descriptions for SRAM subarray. (LBCS: local block column select; LBRS: local block row select)

Design Types	Descriptions
Standard (Std.) Design	FEOL SRAM bitcell array & peripheral logic + BEOL interconnect
Conventional (Conv.) Divided Design	FEOL SRAM bitcell array & peripheral logic + BEOL interconnect + FEOL logic for divided technique
Active Interconnect (AIC) Divided Design	FEOL SRAM bitcell array & peripheral logic + BEOL interconnect + FEOL LBRS and LBCS decoders + BEOL logic for divided technique without LBRS and LBCS decoders

To conduct a comprehensive DTCO analysis involving various subarray options, multiple sizes of subarrays are achieved based on the conventional (Conv.) divided wordline (DWL) [4] and divided bitline (DBL) [5] techniques (Fig. 3(b), Table I, and Fig. 4). For simplicity, the terms “standard (Std.)” and “divided” designs are used to refer to the SRAM subarray with and without DWL and DBL techniques, respectively, in

this work. By implementing selecting and gating circuits to partition the SRAM bitcell array into several local blocks (LBs), the effective RC for BL and WL is decreased. As a result, the Conv. divided subarray design offers significant subarray-level improvements in PP and reduces WFR. However, this improvement comes at the cost of a noticeable subarray area overhead of up to 30% compared to the Std. counterpart (Fig. 5), which may negatively impact the macro-level PPA.

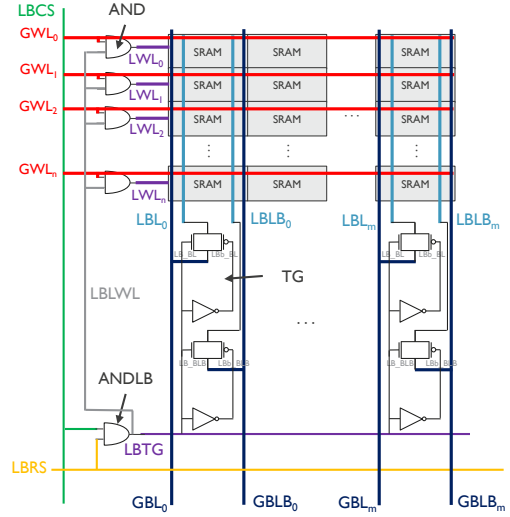


Fig. 4. Circuit diagram of a single LB in divided subarray design for $(WL, BL) = (0, 0)$. (LB: local block; TG: transmission gate; AND: AND gate for LWL; ANDLB: AND gate for LB select; LBRS: local block row select; LBCS: local block column select; GBL/GBLB: global BL/BLB; LBL/LBLB: local BL/BLB; GWL: global WL; LWL: local WL; LBLWL: LWL select in LB)

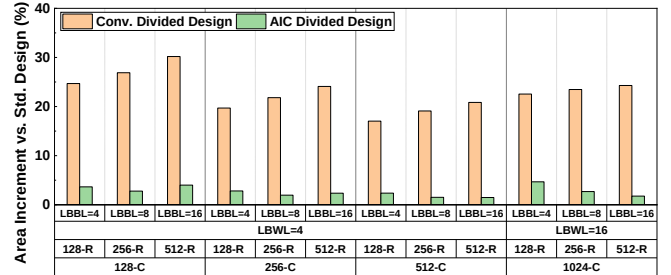


Fig. 5. Subarray area comparison of Conv. and AIC divided subarray designs with the optimized LB in different subarray sizes.

B. Proposed Contribution

To address this subarray area penalty and maintain similar subarray-level PP enhancements, the active interconnect (AIC) technique is proposed for implementation within the divided design (Fig. 3(c) and Table I). The AIC approach shares similarities with the concept of 3-dimensional (3D) sequential non-bonding stacking architecture, which involves a blanket active layer transfer onto prefabricated bottom devices [6, 7, 8, 9, 10, 11, 12, 13]. Note that it is sometimes referred to as monolithic 3D (M3D) integration [11] where multiple layers are fabricated sequentially, which is the opposite of parallel 3D technology based on wafer bonding [6]. 3D stacking technology has emerged as a promising alternative to extend Moore’s law scaling [6], offering increased device density per

chip area and reduced interconnect length. Among 3D techniques, through-silicon-via-based 3D (TSV-3D) stacking is a notable example of parallel 3D technology. However, it faces challenges such as low alignment precision, limited TSV dimension scaling, and taller TSV requirement in vertical direction, resulting in an increase in overall volume [6]. Consequently, 3D sequential (or M3D) technology has gained popularity due to its elimination of alignment issues and the incorporation of fine-grained monolithic inter-via (MIV) connections [6]. To prevent bottom-level device degradation, maintaining low thermal budget of the top-level device and layer transfer processing is crucial, typically below 500°C [7, 8, 9, 10, 11, 12, 13]. The term “BEOL-compatible devices” specifically denotes the top-level device in this context. The key distinction of the AIC technology, compared to existing 3D sequential technology, lies in the insertion of single or multiple layers of active devices within the BEOL, which remains a part of the bottom SRAM subarray. These BEOL devices contribute to the overall SRAM functionality instead of operating as separate logic or memory blocks. Therefore, we use the term AIC rather than 3D sequential to emphasize this unique conceptual framework.

The proposed AIC technology tackles the subarray area penalty issue in the Conv. divided subarray design by integrating BEOL-compatible devices. Several 1- or 2-dimensional (1- or 2D) options, such as carbon-nanotube (CNT, cylinders) [14, 15, 16, 17], MoS₂ [18], and WS₂ [19]-based devices, could be potential candidates. Concerns regarding 2D-based FET [20] arise from higher contact resistance and lower mobility compared to Si FET, which may potentially lead to PP degradation in AIC divided subarray design. Among these devices, Carbon Nanotube FETs (CNFETs) have been projected to achieve large benefits compared to Si FETs [14] due to their superior electrostatic control and large injection velocity [7, 14, 15, 16, 17]. Moreover, CNFETs have already been integrated in the BEOL within multiple commercial silicon fabrication facilities at mature nodes (e.g., 90 nm node) [21, 22], leveraging low-temperature fabrication processes (i.e., <400°C). For these reasons, this work considers gate-all-around CNFETs with high-density CNTs [14] as BEOL devices. By incorporating the feature of fabricating an active region within the SRAM subarray’s BEOL layer, the extra logic gates can be relocated from FEOL to BEOL. Hence, the AIC divided subarray design is predicted to reduce the subarray area overhead from up to 30% to less than 5% (Fig. 5) without any significant degradation in subarray-level PP compared to the Conv. counterparts.

Multiple nanosheet (NS) FET SRAM subarray options are thoroughly optimized across three design types (Std., Conv. divided, and AIC divided designs) with distinct subarray sizes and numbers of local blocks (LBs) in 14-Å-compatible technology (A14) [23]. Furthermore, to validate the value of AIC divided subarray design and motivate more concrete process development, corresponding DTCO PPA analysis [3] is carried out at both subarray and macro levels.

II. SRAM SUBARRAY DESIGNS

This study presents the HD 6T SRAM bitcell array and periphery designs based on A14 NS [23] and A14 CNT [14] technology for devices at FEOL and BEOL, respectively. A quick overview of the three different SRAM subarray designs (Fig. 3 and Table I): Std., Conv. divided, and AIC divided designs, are described in this section. Fig. 6 and Fig. 7 illustrate the schematic views of the single LB layout in conv. and AIC divided designs, respectively, according to the circuit diagram shown in Fig. 4. Their physical parameters for each signal line and the corresponding estimated RC per cell (without device capacitance) extracted by the QuickCap tool [24] are listed in Table II and Table III.

A. Std. design (Fig. 3(a))

This design consists of peripheral logic, including address/timing control, precharge circuits, row decoder, column multiplexer, sense amplifiers, and write drivers, and SRAM bitcell array. All the devices are placed at FEOL. More details of the SRAM subarray architecture can be found in [3]. Expanding the subarray size in the Std. design causes dramatic degradation in RC for BL and WL, resulting in slower speed and write failure. Hence, the available subarray design options become limited.

B. Conv. divided design (Fig. 3(b))

To reduce the effective RC for BL and WL for avoiding write failure, the SRAM bitcell array is divided into several LBs using selecting and gating circuits. The reference layout example is presented in Fig. 6, where every 2 ANDs in sequences are arranged at the same horizontal level starting from the top-left corner. Similarly, every 2 TGs in sequences are placed at the same vertical level starting from the bottom-left corner. Additional logic gates such as AND gates (AND), transmission gates (TG), LB column select (LBSC) decoder, and LB row select (LBRS) decoder are added beyond the Std. design (Fig. 4). Moreover, the isolation gaps (ISO) are required to be inserted in between the logic and SRAM regions. Therefore, the Conv. divided design leads to an area increment of up to 30% among the optimized subarrays compared to the Std. design (Fig. 5). Although the subarray-level PP improves, the overall PP at macro level may be degraded due to the consequent larger inter-subarray interconnect and logics overhead [1, 2].

To achieve the division into several local WL (LWL) and BL (LBL), the ANDs and TGs are connected to global WL (GWL) and BL (GBL), respectively. AND is used to generate the LWL signal with the input of GWL and LBLWL, and TG is the switch between GBL/GBLB and LBL/LBLB. Each LB is activated by selecting the corresponding LBSC and LBRS signals. The number of LBs in WL (LBWL) and BL (LBBL) determines the number of LB in the WL and BL directions, respectively. The number of columns (N_{col_LB}) and rows (N_{row_LB}) in each LB is calculated by dividing the total number of column (N_{col}) and row (N_{row}) in subarray by LBWL and LBBL, respectively. In this initial work, full swing on GBL and LBL is assumed, but there is potential to extend it to full swing GBL and low swing LBL for lower energy designs in the future.

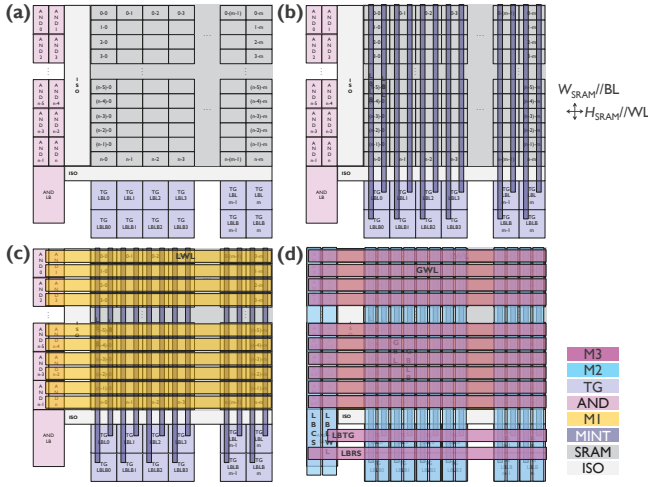


Fig. 6. Schematic views of a single LB layout in Conv. divided subarray design. (a) SRAM bitcell array, isolation gap between SRAM and logic (ISO), AND and TG (FEOL active layer) located around LB. (b) LBL/LBLB (MINT) (c) LWL (M1). (d) GBL/GBLB (M2), GWL (M3), LBCS (M2), LBLWL (M2), LBTG (M3), and LBRS (M3).

Table II. Physical parameters and estimated RC per cell (without device capacitance) for each signal line (ruthenium is assumed) in conventional divided design. (T_{wire} : wire thickness; W_{wire} : wire width; L_{wire} : wire length)

	Metal Layer	T_{wire} (nm)	W_{wire} (nm)	L_{wire} (nm)	R (Ω)	C (aF)
BL	MINT	27	27	84	16	56.6
WL	M1	49	70	150	5.55	177
GBL	M2	27	63	84	6.54	37.3
GWL	M3	49	70	150	5.55	51.3
LBCS	M2	27	81	168	10.08	25.5
LBRS	M3	49	70	150	5.55	26.8
LBLWL	M2	27	81	168	10.08	25.5
LBTG	M3	49	70	150	5.55	26.8

C. AIC divided design (Fig. 3(c))

To overcome the area bottleneck observed in the Conv. divided subarray design, the AIC divided technique is introduced for SRAM subarray. The AIC concept entails the introduction of an additional active layer at BEOL and the integration of BEOL-compatible and low-temperature growth (250~450°C) CNFET [6, 7, 8, 9]. Consequently, it brings significant subarray area savings: less than 5% overhead (Fig. 5) at the FEOL by relocating a portion of the devices to the BEOL layer. It's important to emphasize that CNFET is exclusively utilized for the additional logic gates in the divided design for specific reasons. Firstly, it aids in managing fabrication costs [15, 17], thereby preventing an undue escalation in comparison to the Std. subarray design. Secondly, while several process and design techniques (i.e., imperfection-immune paradigm [25]) have been introduced to mitigate CNFET variability, an assessment of the impact of variability in A14 is still lacking. For example, targets of chiral purification of CNTs (demonstrated in [26, 27, 28, 29] and necessary to achieve low variability in leakage and threshold voltage) are still unclear in A14. As a result, this study predominantly concentrates on integrating CNFET into specific circuit components while excluding SRAM bitcell

array and other peripheral elements. The approach aims to successfully incorporate CNFET in SRAM AIC divided subarray design, avoiding excessively high fabrication costs and operating under less strict variability constraints. Future research should explore hybrid design strategies integrating both Si and CNT technologies in the SRAM bitcell array and its peripheral design. This involves a more in-depth investigation, specifically focusing on mitigating CNFETs variability in A14.

In this approach, all the ANDs and TGs used for the divided techniques are moved to the BEOL active layer and placed on top of the SRAM bitcell array as depicted in the reference layout example (Fig. 7). The arrangement consists of placing every 4 ANDs in sequences at the same horizontal level starting from the top-left corner, and every 2 TGs in sequences at the same vertical level starting from the bottom-left corner. Note that the LBCS and LBRS decoders are retained at the FEOL due to inadequate space available in the BEOL active layer (Fig. 3(c) and Table I).

Fig. 8 and Table III show the schematic views and physical parameters of 3D vias assumed in AIC divided design. The layers from bottom to top include FEOL active region (SRAM and peripheral logics), 1st BEOL (VINT to V2), BEOL active region (AND and TG), and 2nd BEOL (V1 to M7). Notably, the 2nd BEOL initiates from V1 layer due to the pitch requirement of customized layout designs shown in Fig. 7. The term “3D via” represents any type of vias (e.g., stacked via, buried power rail, nanoscale through-silicon via, etc.) that vertically connects BEOL devices to the 1st BEOL layer without additional area cost. For a simple RC approximation of 3D vias, stacked via is assumed for the connection between LBL to TG, TG to GBL, LWL to AND, and AND to GWL without any shift in between (Fig. 8). Accurate RC values should be extracted by a full-custom layout design in the future, which is not the primary focus of the current study.

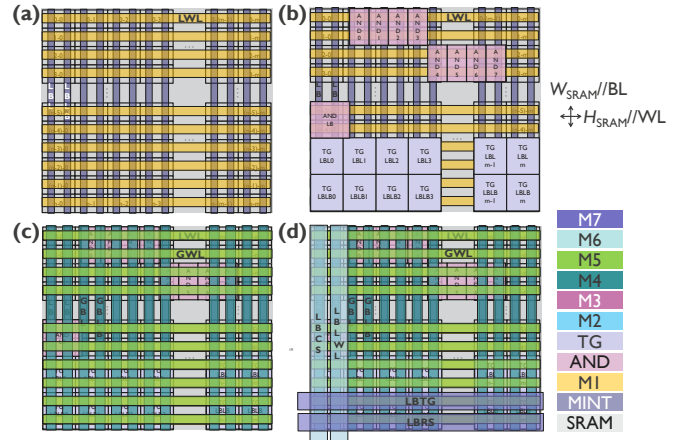


Fig. 7. Schematic views of a single LB layout in AIC divided subarray design. (a) SRAM bitcell array, LBL/LBLB (MINT), and LWL (M1). (b) AND and TG (BEOL active layer). (c) GBL/GBLB (M4) and GWL (M5). (d) LBCS (M6), LBLWL (M6), LBTG (M7), and LBRS (M7).

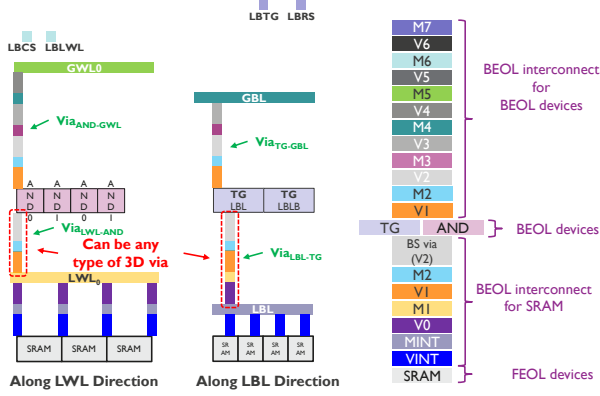


Fig. 8. Schematic views of 3D vias along LWL and LBL directions in AIC divided subarray design.

Table III. Physical parameters and estimated RC per cell (without device capacitance) for each signal line (ruthenium is assumed) and 3D vias in AIC divided design. (T_{wire} : wire thickness; W_{wire} : wire width; L_{wire} : wire length)

	Metal Layer	T_{wire} (nm)	W_{wire} (nm)	L_{wire} (nm)	R (Ω)	C (aF)
BL	MINT	27	27	84	16	56.6
WL	M1	49	42	150	9.45	164
GBL	M4	28	47	84	3.68	22
GWL	M5	28	58	150	5.05	30
LBCS	M6	80	80	168	1.06	26.1
LBRSL	M7	80	80	150	0.95	23.1
LBLWL	M6	80	80	168	1.06	26.1
LBTG	M7	80	80	150	0.95	23.1
$\text{Via}_{\text{BL-TG}}$	V0-V2	16	16	144	88	20.8
$\text{Via}_{\text{TG-GBL}}$	V1-V3	16	16	152	84	19.6
$\text{Via}_{\text{LWL-AND}}$	V1-V2	16	16	75	46	22.7
$\text{Via}_{\text{AND-GWL}}$	V1-V4	16	16	208	101	26.8

D. SRAM Bitcell Layout Design

Fig. 9 displays the three layout designs of HD A14 6T NSFET SRAM bitcell for Std. [8], Conv. divided, and AIC divided subarray designs, each with distinct signal routing requirements. In all designs, the ground voltage (V_{SS}) is assumed to be placed at the backside using buried power rail (BPR) technique [30, 31, 32]. The internal nodes (Q and QB), supply voltage (V_{DD}), and local signals (BL/LBL, BLB/LBLB, and WL/LWL) are located at the MINT and M1 layers. The main difference between the Conv. and AIC designs is the width of LWL, GBL, and GBLB due to distinct pitch limitations (Table II and Table III). In Conv. divided design (Fig. 9(b)), the global signals (GWL and GBL) are connected from the AND and TG located outside the SRAM bitcell regions (Fig. 6). This arrangement leads to M1, M2, and M3 being exclusively occupied by LWL, GBL/GBLB, and GWL, respectively. However, in AIC divided subarray design (Fig. 9(c)), the AND and TG are placed right above the SRAM bitcell regions (Fig. 7). Therefore, the metal widths in AIC divided subarray design become narrow since extra spaces are required for the vertical connections from GBL/GBLB and GWL to the bitcell compared to the conventional counterpart.

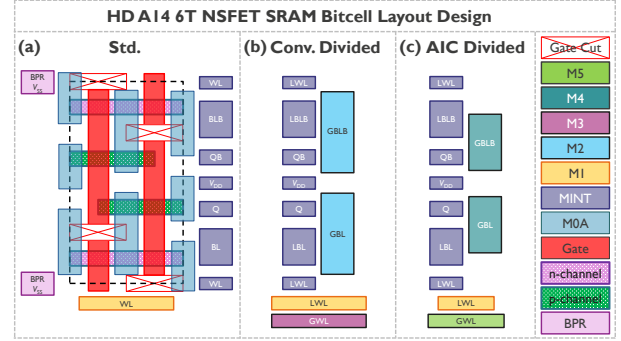


Fig. 9. Schematic views of HD A14 6T NSFET SRAM bitcell layout design for (a) Std., (b) Conv. divided, and (c) AIC divided subarray designs.

III. CNFET OPTIMIZATION IN AIC DIVIDED SUBARRAY DESIGN

A14 CNFET has demonstrated the ability to deliver the I_{on} that is three times as large as that of A14 Si NSFET at $V_{\text{DD}} = 0.45\text{V}$ and off current ($I_{\text{off}} = 2\text{ nA}$) [14]. This higher current capability of CNFET may help offset the additional parasitic RC introduced by the AIC divided subarray design. Therefore, CNFETs are introduced in AIC divided subarray design for further PPA analysis. In this work, the model proposed in [14] was utilized, including both nanosheet (NS) and forksheet (FS) based layouts (NS and FS CNFET). The simulations were conducted utilizing Cadence Virtuoso [33] and Cadence Spectre [34].

Table IV. Optimized physical parameters for A14 Si NSFET, A14 CNFET, and N22 FDSOI devices in AIC divided subarray design at $V_{\text{DD}} = 0.7\text{V}$. (NS: nanosheet; FS: forksheet; CPP: contacted poly pitch; L_{G} : gate length; W_{ch} : channel width; NNS: number of nanosheet stacks; d_{CNT} : CNT diameter; p_{CNT} : CNT pitch; I_{on} : on current, I_{off} : off current)

Parameters	A14 Si NSFET		A14 CNFET		N22 FDSOI
	SRAM	Logic	NSFET	FSFET	Planar
CPP (nm)	42				104
L_{G} (nm)	15	14	14	14	20
W_{ch} (nm)	11	12	12	21	170
NNS	4	4	3	2	--
d_{CNT} (nm)	--	--	0.85	0.85	--
p_{CNT} (nm)	--	--	2	2	--
V_{DD} (V)	0.7				0.8
I_{on} ($\mu\text{A}/\text{FET}$)	69	91	164	199	113
I_{off} (nA/FET)	0.02	0.2	0.2	0.2	0.2
Model Ref.	[23]		[14]		[37]

A. Model Calibration

In this section, the simulations are performed at nominal $V_{\text{DD}} = 0.7\text{V}$, temperature = 27°C , and typical-typical (TT) corner. Except for the SRAM bitcell, where the I_{off} is assumed to be 0.02 nA , the assumption of $I_{\text{off}} = 0.2\text{ nA}$ for all the peripheral logics remains identical in both Si and CNFETs models. The optimization of CNFETs is conducted considering their distinct applied voltage, on current (I_{on}), subarray sizes, and application in comparison to the previous work [14]. To achieve the optimal read delay and write margin while maintaining identical I_{off} between CNFET and Si NSFET models, the number of stacked nanosheets (NNS) of CNFET (NNS_{CNT}) and

CNT diameter (d_{CNT}) were systematically optimized. Table IV listed the optimized results of physical parameters for the devices in AIC divided design. After completing this model optimization, the subarray- and macro-level DTCO are analyzed in the following section. It is important to note that the LB sizes used here have been optimized for each size of subarray as discussed in the next section.

a. Impact of NNS_{CNT}

Fig. 10 and Fig. 11 show the impact of the NNS_{CNT} on read delay (RD) and write margin (WM) in the AIC divided subarray design with various subarray sizes. These simulations are taken under fixed NS width (W_{NS}) = 12 nm, d_{CNT} = 0.8 nm, CNT pitch (p_{CNT}) = 2 nm, and NNS of Si NSFET (NNS_{Si}) = 4. The read and write metrics are determined by the balance between the increased strength of the transistors and the larger device capacitance as NNS increases [17]. As a result, the optimized NNS is found to be 3 and 2 for NS and FS CNFET, respectively, as values tend to saturate beyond these NNS (Table IV).

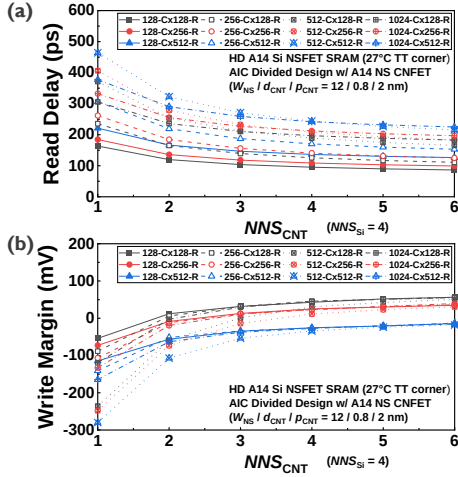


Fig. 10. (a) RD and (b) WM with varying NNS_{CNT} of A14 NS CNFET ($W_{\text{NS}} / d_{\text{CNT}} / p_{\text{CNT}} = 12 / 0.8 / 2$ nm) for HD A14 Si NSFET ($W_{\text{NS}} = 12$ nm; $NNS_{\text{Si}} = 4$) AIC divided subarray design.

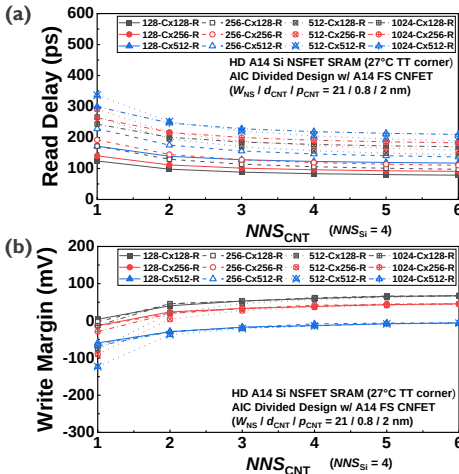


Fig. 11. (a) RD and (b) WM with varying NNS_{CNT} of A14 FS CNFET ($W_{\text{NS}} / d_{\text{CNT}} / p_{\text{CNT}} = 21 / 0.8 / 2$ nm) for HD A14 Si FSFET ($W_{\text{NS}} = 21$ nm; $NNS_{\text{Si}} = 4$) AIC divided subarray design.

b. Impact of d_{CNT}

Fig. 12 and Fig. 13 demonstrate the impact of d_{CNT} on read delay and write margin in the AIC divided subarray design with various subarray sizes. These simulations are performed with fixed values of $W_{\text{NS}} = 12$ nm, $p_{\text{CNT}} = 2$ nm, and $NNS_{\text{Si}} = 4$. The sweeping range for d_{CNT} is from 0.7 to 0.89 nm, ensuring the same leakage assumption (0.2 nA) [35]. As d_{CNT} increases, both the transistor's strength and device capacitance increase. However, saturation or degradation occurs in RD and WM when d_{CNT} exceeds 0.85 nm due to the increased device capacitance. Thus, an optimized value of d_{CNT} at 0.85 nm is considered suitable for both NS and FS CNFETs (Table IV).

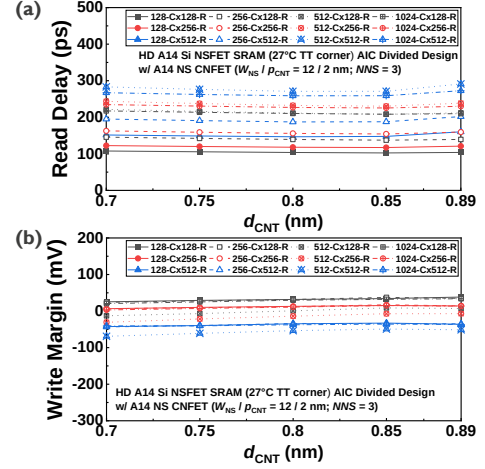


Fig. 12. (a) RD and (b) WM with varying d_{CNT} of A14 NS CNFET ($W_{\text{NS}} / p_{\text{CNT}} = 12 / 2$ nm; $NNS_{\text{CNT}} = 3$) for HD A14 Si NSFET ($W_{\text{NS}} = 12$ nm; $NNS_{\text{Si}} = 4$) AIC divided subarray design.

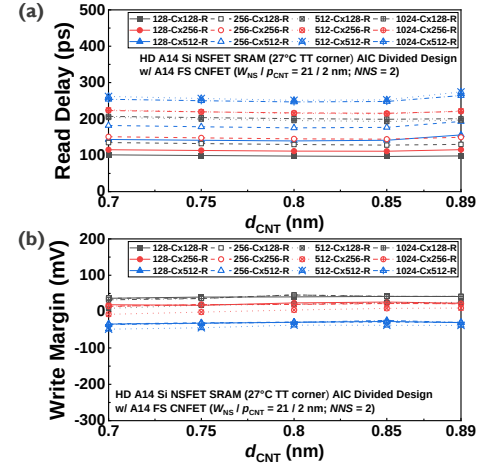


Fig. 13. (a) RD and (b) WM with varying d_{CNT} of A14 FS CNFET ($W_{\text{NS}} / p_{\text{CNT}} = 21 / 2$ nm; $NNS_{\text{CNT}} = 2$) for HD A14 Si NSFET ($W_{\text{NS}} = 12$ nm; $NNS_{\text{Si}} = 4$) AIC divided subarray design.

B. Variability

Several key factors such as the variation in d_{CNT} , chirality (yield of metallic or semiconducting CNTs, density, bandgap), and doping, play a vital role in shaping the characteristics of CNFET [17]. These sources of variation add to the traditional variability encountered in silicon CMOS, which may affect the

overall PP results. While existing process and design techniques for mitigating these variations were well-demonstrated [16, 27, 28, 29], CNT pitch uniformity and CNT alignment control at the wafer-scale still represent a key challenge. However, this discussion goes beyond the primary focus of this work. It is imperative to extend the investigation of the variation's impact on CNFETs associated with the SRAM AIC divided subarray design in future studies.

IV. SUBARRAY OPTIMIZATION FOR HD OPTION

A. Benchmark at Subarray Level

The optimized design types and their corresponding LB sizes for each subarray are summarized in Table V. Note that these LB sizes have undergone optimization across all the LBWL and LBBL configurations based on the write-ability studies shown in Fig. 14 [3]. Fig. 15, Fig. 16, Fig. 17, and Fig. 18 present the comparison for various subarray row (-R) and column (-C) with optimized LB sizes between Std., Conv. divided, and AIC divided subarray designs at both subarray and macro levels.

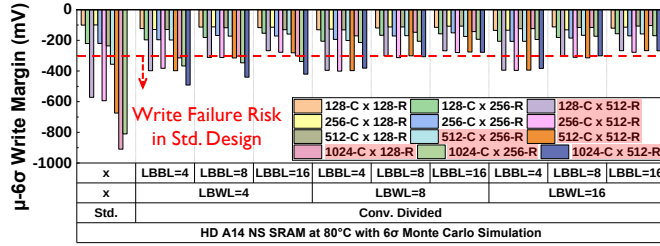


Fig. 14. Worst-case WM of HD A14 NSFET SRAM with Conv. divided subarray design based on 6-sigma (6σ) Monte Carlo simulation at the worst-case temperature (80°C). Subarrays that experience write operation failures are highlighted in red color.

For the subarray sizes of 128-C/256-C $\times$ 128-R (Fig. 15), the Std. design demonstrates similar RD, WM, and write delay (WD) as the Conv. and AIC divided subarray designs without compromising the subarray area. As a result, the Std. subarray design is considered the optimized design type for 128-C/256-C $\times$ 128-R. Notably, the comparable margin and performance observed in smaller subarrays between Si and CNT models can be attributed to the compensatory effect of the delay overhead from additional logics' RC even though CNFET has relatively higher I_{on} than Si NSFET in A14. However, for the remaining N_{row} and N_{col} configurations, the Conv. and AIC divided subarray designs outperform the Std. counterpart. Hence, although the Std. subarray designs for 256-C $\times$ 128/256-R and 512-C $\times$ 128-R configurations do not suffer from WFR, they are not regarded as optimized designs. Particularly, in the case of 512-R, the Std. subarray design encounters write failures necessitating the divided techniques to ensure successful write operations (Fig. 14). By utilizing Conv. divided technique, the WM and RD improve by 338 mV and 37%/68%, respectively, for 512-C/1024-C $\times$ 512-R. Among all the designs, the AIC divided subarray design stands out as it offers the best performance and write margin with only a slight increase in area of less than 5% compared to the Std. counterpart (Fig. 5). AIC divided subarray design even provides additional 99 mV WM and 37% RD enhancements compared to Conv.

counterparts due to faster CNFET. This promising result prompts further PPA exploration at the macro level to validate its potential value.

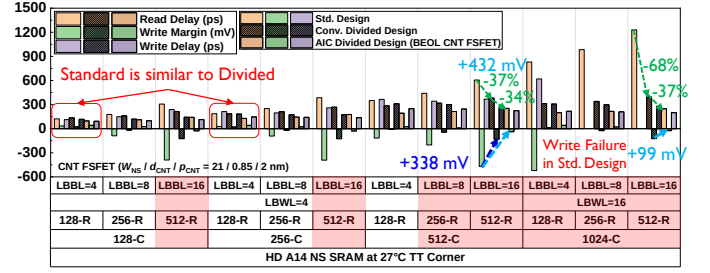


Fig. 15. SRAM subarray-level RD, WM, and WD benchmark in the Std., Conv. divided, and AIC divided subarray designs with optimized LBWL and LBBL for each subarray size. Subarrays that experience write operation failures are highlighted in red color.

Table V. Optimized (opt.) results of each subarray with the corresponding LB size between Std. and divided subarray designs.

N_{col}	N_{row}	Opt. Design	LBWL	LBBL
128-C	128-R	Standard	$\times$	$\times$
	256-R	Divided	4	8
	512-R	Divided	4	16
256-C	128-R	Standard	$\times$	$\times$
	256-R	Divided	4	8
	512-R	Divided	4	16
512-C	128-R	Divided	4	4
	256-R	Divided	4	8
	512-R	Divided	4	16
1024-C	128-R	Divided	16	4
	256-R	Divided	16	8
	512-R	Divided	16	16

B. Benchmark at Data Array Macro Level

Fig. 16 and Fig. 17 present macro-level simulation data based on a 16 MB data capacity, center pin access, and H-tree interconnect technology [1, 2]. As shown in Fig. 16 and Fig. 17, a notable 11%, 23%, 36%, and 56% of delay, energy, area, and EDAP gain, respectively, can already be achieved in Std. design by simply transitioning from small (128-C $\times$ 128-R) to large (1024-C $\times$ 512-R) subarray size due to less H-tree contribution (disregarding WFR). Fig. 16 clearly shows that the inter-subarray interconnect has a decisive influence on the overall macro-level PPA. To optimize this aspect, choosing a bigger-size subarray is the preferred approach to reduce the interconnect overhead. Further macro-level PPA improvement is still achievable even if they involve a larger subarray-level PPA overhead, which is relatively less impactful than inter-subarray interconnect. It is worth noting that increasing N_{row} is more effective for PP improvement compared to N_{col} since the longest wire is on horizontal direction [1, 2, 36].

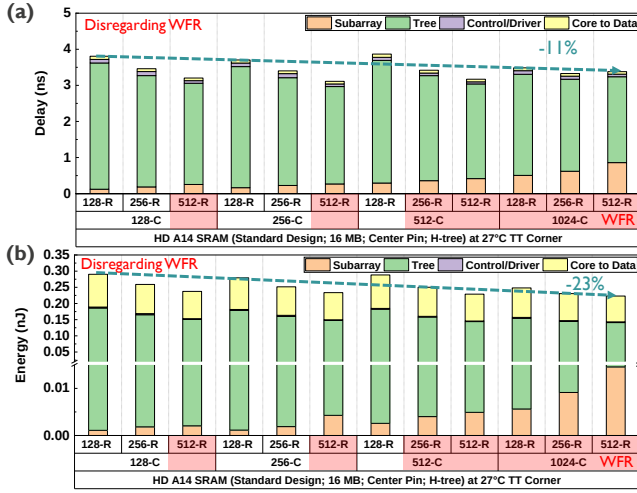


Fig. 16. SRAM data macro-level (a) delay and (b) energy breakdown in the Std. subarray designs (disregarding WFR) for each subarray size with data capacity of 16 MB, **center pin access**, and H-tree technology. Data arrays that experience write operation failures are highlighted in red color.

While employing a bigger-size subarray with Std. subarray design may suffer from the WFR, this problem can be effectively addressed through the implementation of divided design. In the previous subsection, the Conv. divided subarray design substantially enhances subarray-level performance and WM compared to the Std. counterpart (Fig. 15). However, this enhancement comes at the cost of unignorable subarray-level area penalty that could potentially lead to larger inter-subarray interconnects and logic gates overhead at macro level. Consequently, this 24% macro-level area overhead in Conv. divided subarray design causes 26% macro-level EDAP degradation compared to Std. counterpart (Fig. 17). To address these challenges, AIC divided subarray design has been proposed to mitigate the macro-level area overhead to less than 3% (Fig. 17 (a)). AIC divided subarray design also successfully provides up to 65% macro-level EDAP improvement compared to a realistic Std. counterpart that does not have WFR (Fig. 18). Fig. 18 indicates the preferred selection between AIC divided subarrays, either with a subarray size of 1024-C×256-R or 1024-C×512-R. This decision stems from the negligible EDAP gain (0.2%) in a 1024-C×128-R subarray compared to the Std. 256-C×256-R subarray. Even assuming a non-failing write in Std. subarray design, AIC divided subarray design still achieves a notable 19% enhancement with a 1024-C×512-R subarray size (Fig. 17(b)). Notably, the increased macro-level area in the AIC divided subarray design associated with CNFET fabrication may cause a higher overall die cost when compared to the Std. counterpart and the traditional Si FET process. A more comprehensive cost analysis is advisable for future research, which falls outside the central scope of this work.

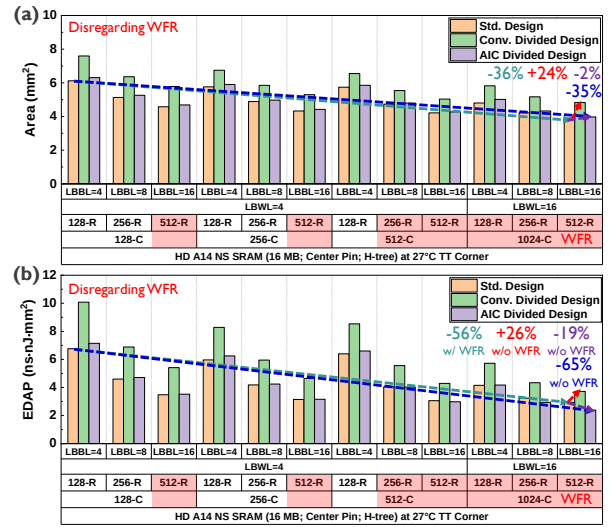


Fig. 17. SRAM data macro-level (a) area and (b) EDAP benchmark in the Std. (disregarding WFR), Conv. divided, and AIC divided designs with optimized LBWL and LBBL for each subarray size with data capacity of 16 MB, **center pin access**, and H-tree technology. Data arrays that experience write operation failures are highlighted in red color.

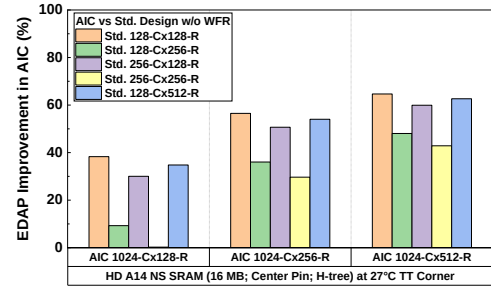


Fig. 18. EDAP improvements in AIC divided of 1024-C×128/256/512-R compared to Std. design (regarding WFR) for subarray size of 128/256-C×128-R, 128/256-C×256-R, and 128-C×512-R with data capacity of 16 MB, **center pin access**, and H-tree technology.

C. Comparison of Si, FDSOI, and CNT models

In this section, the examination focuses on the comparison of utilizing A14 Si NSFET [23], 22-nm (N22) fully depleted silicon-on-insulator (FDSOI) [37], and A14 CNFET [14] models for realistic SRAM AIC divided subarray design to highlight the benefits of employing CNFET devices (Fig. 19 and Fig. 20). Over the past few years, several experimental studies on BEOL-compatible and low-temperature-fabricated CNFETs for 3D integration have been reported [6, 7, 8, 9], establishing them as realistic candidates for further 3D integrated-circuit (IC) development. Additionally, investigations into low-temperature-fabricated FDSOI as top-level transistors (below 500°C) for 3D stacking technology have been conducted including electrical characteristics and reliability measurements [10, 12, 13]. However, it's important to note that the FDSOI device is a planar-type transistor, with mass production limited to 22 nm by GlobalFoundries until now [37]. Apparently, the device dimensions of N22 FDSOI transistors could not align with A14 design rules in this study.

Therefore, the FEOL-compatible A14 Si NSFET model (identical model as SRAM bitcell in this work) [23], regardless of potential degradation from the low-thermal-budget process, is also considered as an optimistic upper bound for BEOL logics in our realistic AIC technology comparison.

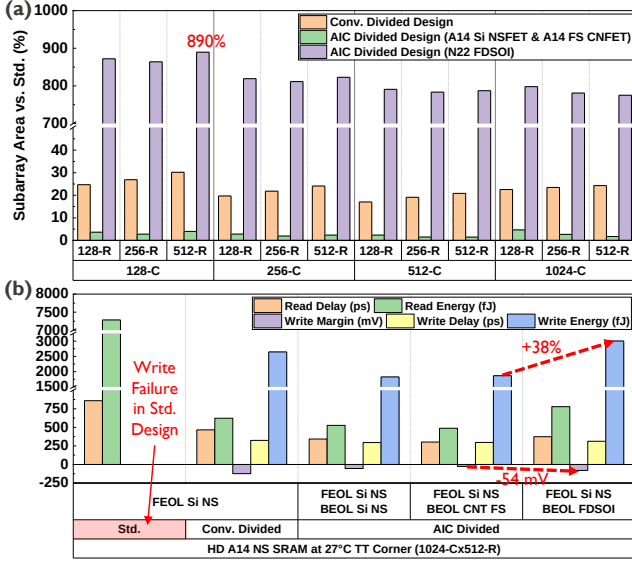


Fig. 19. SRAM subarray-level (a) area, (b) margin, performance, and power comparison in Std., Conv. divided, and AIC divided subarray designs assuming A14 Si, A14 CNT, and N22 FDSOI models for BEOL logics. Subarrays that experience write operation failures are highlighted in red color.

Fig. 19 and 20 show the subarray- and data array macro-level PPA benchmark, respectively, across five distinct subarray designs: Std., Conv. divided, AIC divided with A14 Si NSFET, AIC divided with A14 FS CNFET, and AIC divided with N22 FDSOI. The parameters for the N22 FDSOI device, including V_{DD} , gate length, width, contacted poly pitch, and metal pitch, are set at 0.8 V, 20 nm, 170 nm, 104 nm, and 80 nm, respectively, based on low V_T (LVT) transistors from GlobalFoundries' 22FDXTM technology (Table IV) [37]. However, the implementation of the AIC divided subarray design with the N22 FDSOI model results in a considerable subarray area overhead, reaching up to 890% compared to the Std. counterpart (Fig. 19(a)). This amplified area incurs substantial macro-level PPA degradation due to longer inter-subarray interconnects (Fig. 20). Moreover, the N22 FDSOI model shows a 54 mV decrease in write margin and a 38% increase in write energy at the subarray level (1024-C×512-R) in comparison to the A14 FS CNFET counterparts, attributed to higher V_{DD} and inferior device performance (Fig. 19(b)). Consequently, while the N22 FDSOI represents an existing solution for 3D integration, it proves unsuitable and serves as a pessimistic lower bound for the AIC divided subarray design in the ultra-scaled A14 technology. This result indicates that BEOL-compatible logic still has strict requirements in terms of logic density.

Conversely, the AIC divided subarray design with A14 Si NSFET model demonstrates comparable (less than 2% difference) performance/power and identical area results to the

A14 FS CNFET model at both subarray and macro levels. This validates the inherent advantages of employing CNFET as BEOL-compatible devices in the AIC divided subarray design, surpassing the optimistic yet unfeasible upper bound showcased by the A14 Si NSFET counterpart.

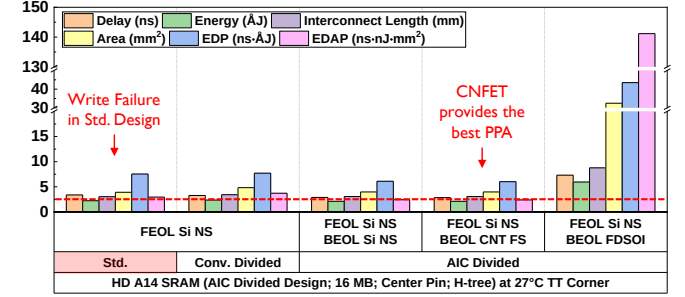


Fig. 20. SRAM data macro-level benchmark comparison in Std., Conv. divided, and AIC divided subarray designs assuming A14 Si, FDSOI, and A14 CNT models for BEOL logics. Data arrays that experience write operation failures are highlighted in red color. (EDP: energy-delay product)

V. CONCLUSION

This study presents a comprehensive DTCO analysis, establishing the potential of the AIC divided subarray design integrated with CNFET as BEOL-compatible devices for future PPA improvements of 3D SRAM data array design. In interconnect-dominated nodes, macro-level PPA is heavily influenced by the inter-subarray interconnect, a factor intricately linked to the subarray size. By scaling up the subarray size from 128-C×128-R to 1024-C×512-R, remarkable 11%, 23%, 36%, and 56% of macro-level delay, energy, area, and EDAP improvements, respectively, are achieved, albeit with the trade-off of a WFR in the Std. subarray design. To address this WFR, the Conv. divided subarray design has been explored, offering an improved WM of 338 mV. However, this enhancement comes at the cost of up to 30% additional macro-level area overhead. To circumvent this area penalty and maintain it below 3%, the AIC divided technique emerges as a promising solution for SRAM subarray design. It not only further enhances the WM (+99 mV) compared to Conv. counterparts but also provides up to 65% macro-level EDAP improvement compared to the Std. counterpart, which can only employ a relatively smaller but realistic subarray size. This underscores the potential value of the AIC divided subarray design in advancing PPA of SRAM cache.

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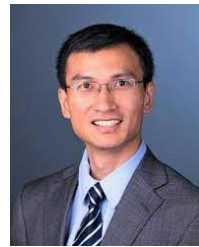
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